



Dimensions of data analytics

Hi. I'm Cassie, and I lead

Decision Intelligence for Google Cloud.

Decision Intelligence is a combination of

applied data science and

the social and managerial sciences.

It is all about harnessing the power and beauty of data.

I help Google Cloud and its customers turn

their data into impact and

make their businesses and the world better.

A data analyst is an explorer,

a detective, and an artist all rolled into one.

Analytics is the quest for inspiration.

You don't know what's going to

inspire you before you explore,

before you take a look around.

When you begin, you have no idea what

you're going to find and

whether you're even going to find anything.

You have to bravely dive into the

unknown and discover what lies in your data.

There is a **pervasive** myth that
someone who works in data
should know the everything of data.

I think that that's unhelpful because
the universe of data has expanded.

It's expanded so much
that specialization becomes important.

It's very, very difficult for
one person to know and be the everything of data.

That's why we need these different roles.

The advice that I give folks who are entering the space
is to pick their specialization based on which flavor,
which type of impact best suits their personality.

Now, data science, the discipline of making data useful,
is an umbrella term that **encompasses** three disciplines:
machine learning, statistics, and analytics.

These are separated by
how many decisions you know
you want to make before you begin with them.

**If you want to make a few important decisions under
uncertainty, that is statistics.**

**If you want to automate, in other words, make many, many,
many decisions under uncertainty,
that is machine learning and AI.**

But what if you don't know
how many decisions you want to make before you begin?

What if what you're looking for is inspiration?

You want to encounter your unknown unknowns.

You want to understand your world.

That is analytics.

When you're considering data science and
you're choosing which area to specialize in,
I recommend going with your personality.

Which of the three excellences
in data science feels like a better fit for you?

The excellence of statistics is rigor.

Statisticians are essentially philosophers,
epistemologists.

They are very, very careful about
protecting decision-makers from coming
to the wrong conclusion.

If that care and rigor is what you are passionate about,
I would recommend statistics.

Performance is the excellence of
the machine learning and AI engineer.

You know that's the one for you if someone says to you,
"I bet that you couldn't build
an automation system that performs this task with
99.99999 percent accuracy," and
your response to that is, "Watch me."

How about analytics?

The excellence of an analyst is speed.

How quickly can you surf through vast amounts of data to explore it and discover the gems, the beautiful potential insights that are worth knowing about and bringing to your decision-makers?

Are you excited by the **ambiguity** of exploration?

Are you excited by the idea of working on a lot of different things, looking at a lot of different data sources, and thinking through vast amounts of information, while promising not to **snooze** past the important potential insights?

Are you okay being told,

"Here is a whole lot of data.

No one has looked at it before.

Go find something interesting"?

Do you thrive on creative, open-ended projects?

If that's you, then

analytics is probably the best fit for you.

A piece of advice that I have

for analysts getting started on

this journey is it can

be pretty scary to explore the unknown.

But I suggest letting go a little bit of

any temptations towards perfectionism and instead,

enjoying the fun, the thrill of exploration.

Don't worry about right answers.

See how quickly you can unwrap this gift
and find out if there is anything fun in there.
It's like your birthday, unwrapping a bunch of things.
Some of them you like. Some of them you won't.
But isn't it fun to know what's actually in there?